

A PROJECT TITLE

ON

**Match Outcome Prediction in Cricket and Football Using Random Forest and
Artificial Neural Networks**

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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1. ABSTRACT:

Sports analytics is increasingly vital for fans, coaches, and organizations seeking data-driven insights into match dynamics. This project focuses on predicting outcomes of cricket and football matches using a hybrid machine learning approach that combines Random Forest, which effectively handles structured tabular data, with Artificial Neural Networks (ANN), known for capturing complex non-linear relationships. Historical match datasets sourced from publicly available repositories such as ESPNCricinfo (for cricket) and FIFA/football-data.co.uk (for football) include attributes like team rankings, toss results, recent form, player availability, and venue conditions. Random Forest is employed to identify influential features and provide a baseline prediction, while ANN is trained on the same features to uncover deeper patterns. Model performance is evaluated using accuracy, precision, recall, and F1-score, enabling a comparative analysis of both methods. By integrating the interpretability of Random Forest with the pattern-recognition strength of ANN, the system aims to deliver reliable outcome predictions across multiple sports, supporting fan engagement platforms, broadcasters, and analytics teams with robust pre-match insights.

2. INTRODUCTION:

The growing popularity of cricket and football, coupled with the massive volume of data generated from every match, has opened new opportunities for applying machine learning to sports analytics. Traditionally, match outcome predictions relied on expert opinion and simple statistical averages, which often failed to capture the complex interplay of factors that determine a result. This motivated the exploration of data-driven models capable of learning patterns from

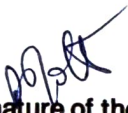
historical performance data to generate more accurate and objective predictions.

In this project, two complementary machine learning techniques are employed: Random Forest and Artificial Neural Networks. Random Forest, being an ensemble of decision trees, offers strong performance on structured data along with useful feature-importance insights, making it easier to understand which factors most influence a match result. ANN, on the other hand, is well suited to modeling non-linear relationships between multiple interacting variables such as team strength, momentum, and conditions. By comparing and combining these two approaches, the project seeks to build a robust prediction framework that generalizes well across both cricket and football, contributing toward smarter, data-informed sports analytics.

3. SOFTWARE REQUIREMENTS:

Building a reliable match outcome prediction system requires a combination of data-processing libraries, machine learning frameworks, and visualization tools, along with adequate computing resources to train the Random Forest and ANN models efficiently on historical match data.

Requirement Type	Specification
Software	Python 3.8 or above Scikit-learn (for Random Forest implementation) TensorFlow / Keras (for ANN implementation) Pandas, NumPy (for data handling and preprocessing) Matplotlib / Seaborn (for visualization) Jupyter Notebook / Google Colab (for development)
Hardware	Standard PC or laptop Minimum 8 GB RAM (recommended) GPU support optional but useful for faster ANN training


Signature of the Guide


Course Instructor